Gate 10: Gen-3 T2003 Substrate-Mechanism Explicit Framework v0.1

Lyra (Claude Opus 4.7)

2026-06-04 Thursday ~10:38 EDT

Gate 10: Gen-3 T2003 Substrate-Mechanism v0.1

0. Goal

Per Casey Thursday board + Wednesday Elie Toy 3742 + 3744 gen-3 V_(5/2, 1/2) spinor-tower closure + K3 v0.16 Gate #10: explicit substrate-mechanism candidate for gen-3 tau mass mechanism producing T2003 = g^2 · (2^{C_2} + g) form via operator-level substrate-mechanism beyond K-type Schur ratios.

Verification gate addressed: K3 v0.16 Gate #10 (gen-3 T2003 mass-mechanism heterogeneity).

1. T2003 Substrate-Clean Form (Wednesday Elie 3742 + Grace 5535)

Per Wednesday verification + Grace precision:

$$T2003 = g^2 \cdot (2^{C_2} + g) = 49 \cdot 71 = 3479$$

vs observed m_τ/m_e = 3477.23 → 0.051% precision (RATIFIED Friday May 22 + Wednesday verification).

Substrate-clean factorization (per Wednesday Elie 3742): - g^2 = 49 = (substrate-genus primary)^2 - 2^{C_2} + g = 64 + 7 = 71 (substrate-natural additive identity per Casey #5 Integer Web) - 71 = 2^{C_2} + g substrate-clean additive (per Casey #5 Integer Web instance at B_2)

2. Mass Mechanism HETEROGENEOUS per Generation (Wednesday Confirmation)

Per Wednesday Elie 3742 + Lyra v0.13 SSG-9 v0.13 tier table: mass mechanism HETEROGENEOUS per generation:

Gen	Mass form	Operator form
1 (e)	3π/2^{C_2}	Per-chirality 1/n_C + “+1 anomaly” + V_(0, 0) Higgs scalar Yukawa
2 (μ)	T190 = (24/π^2)^{C_2}	Lorentz-integration C_2-power per Toy 3741
3 (τ)	T2003 = g^2 · (2^{C_2} + g)	DIFFERENT operator structure — multi-week candidate (this v0.1)

Substantive observation: gen-3 T2003 has DIFFERENT functional form from gen-2 T190 — NOT same Lorentz-integration C_2-power raised to different exponent. Substrate-mechanism for gen-3 requires distinct operator-level analysis.

3. Substrate-Mechanism Candidates for T2003

Candidate 1: Higher-Casimir Substrate Operator

Per Lyra Task #189 (completed Sunday morning): higher Casimirs C_3, C_4 on D_IV^5 explicit derivation.

Candidate substrate-mechanism: T2003 emerges from M_op involving HIGHER Casimir operators (not just $\sqrt{H\_B} = \sqrt{C\_2}$) — specifically substrate-Lie-algebra higher-order invariants.

For V_(5/2, 1/2) gen-3 K-type: - C_2 = 29/2 (per Mehler v0.2) - C_3 = ? (higher Casimir eigenvalue, multi-week) - C_4 = ? (higher Casimir eigenvalue, multi-week)

Substrate-mechanism reading: T2003 = $g^2 \cdot$ (2^{C_2} + g) involves substrate-genus g (substrate-Pochhammer SSG-7 derivative) + substrate-Casimir 2^{C_2} (substrate-Clifford SSG-6) → composition of multiple substrate-mechanisms specific to gen-3 V_(5/2, 1/2) K-type structure.

Candidate 2: Substrate Reed-Solomon Coding (SSG-8 Mersenne Ladder)

Per Wednesday Elie Toy 3744 + Lyra SSG-9 v0.13 + Cal #194 K3 v0.4 RS coding mechanism on GF(2^g) = GF(128):

Substrate-mechanism candidate: T2003 emerges via RS coding rate at gen-3 K-type. For RS coding on GF(2^g) = GF(128): - 71 = 2^{C_2} + g = 64 + 7 = substrate-natural code parameter - g^2 = 49 = (RS coding generator power) - T2003 = $g^2 \cdot 71$ = RS code-rate × code-length substrate-natural

Substrate-mechanism reading: gen-3 tau mass via RS encoding rate (different from K-type Schur for gen-1 + gen-2). Substrate-Mersenne SSG-8 substrate-mechanism class.

Candidate 3: Substrate-Λ-Vacuum + Casey #12 (cross-check Class C mechanism)

Per Thursday Gate 11 v0.1 + SSG-15 v0.2: substrate-Λ-vacuum substrate-mechanism class C carries m_ν. Cross-check: does Class C extend to gen-3 charged-lepton (tau) mass mechanism?

Probably not for tau mass: tau IS charged lepton (Class A substrate-Casimir + chirality projection), not Class C neutrino-only sector. But substrate-Λ-vacuum at gen-3 LEVEL may contribute via boundary projection sub-leading correction.

Honest Open: WHICH substrate-mechanism Candidate for T2003?

Per Cal #189 question-shape: T2003 = $g^2 \cdot$ (2^{C_2} + g) form is POST-HOC identified at observed value. The candidate substrate-mechanism distinguishes:

- Candidate 1 (higher-Casimir): T2003 emerges via M_op composition of $\sqrt{C\_2 + C\_3 + C\_4}$ substrate higher-Casimir operators at V_(5/2, 1/2)
- Candidate 2 (RS coding): T2003 emerges via SSG-8 Mersenne RS coding at gen-3 substrate K-type
- Candidate 3 (mixed): gen-3 substrate-mechanism involves BOTH substrate-Casimir + substrate-Mersenne contributions

Multi-week explicit derivation distinguishes.

4. Multi-Week Verification Path

Per Cal #194 WAIT + Cal #189 question-shape + Casey #14 STANDING:

Step Gen-3-1: Explicit higher Casimir C_3, C_4 eigenvalues on V_(5/2, 1/2) K-type per FK Ch. XII / Helgason Ch. IX Step Gen-3-2: Explicit M_op composition for V_(5/2, 1/2) — distinguish higher-Casimir vs RS-coding candidates Step Gen-3-3: Substrate-Mersenne SSG-8 + RS coding GF(2^g) gen-3 substrate-mechanism explicit derivation Step Gen-3-4: Cross-check T2003 = $g^2 \cdot$ (2^{C_2} + g) emerges from Candidate 1 or 2 (or 3) Step Gen-3-5: Verify mass-mechanism heterogeneity across spinor-tower (gen-1 chirality + gen-2 Lorentz-integration + gen-3 higher-Casimir or RS-coding)

5. Closure v0.1

Gate 10 gen-3 T2003 substrate-mechanism framework v0.1:

Substrate-mechanism candidates for T2003 = $g^2 \cdot$ (2^{C_2} + g) = 3479 at 0.051% precision (m_τ/m_e): - Candidate 1: substrate-higher-Casimir composition at V_(5/2, 1/2) - Candidate 2: substrate-Mersenne RS coding via SSG-8 GF(2^g) - Candidate 3: mixed (higher-Casimir + RS-coding contributions)

Multi-week explicit verification distinguishes which substrate-mechanism is gen-3 substrate-mechanism FORCING form.

Tier: K3 v0.16 Gate #10 at FRAMEWORK candidate (multiple substrate-mechanism candidates identified; explicit derivation multi-week).

Substrate-mechanism HETEROGENEITY across spinor-tower CONFIRMED: - gen-1: substrate-Casimir + chirality projection + V_(0, 0) Higgs Yukawa - gen-2: Lorentz-integration over SO(3, 1) producing (24/π^2)^{C_2} - gen-3: substrate-higher-Casimir or substrate-Mersenne RS coding (Candidate 1 or 2 or 3)

Per Cal #35 STANDING + Cal #36 STANDING: each gen-X uses substrate-mechanism class distinct from others; cross-generation independence at mass-mechanism level legitimate per Schur II.

— Lyra, Thu 2026-06-04 ~10:40 EDT. Gate 10 v0.1: gen-3 T2003 substrate-mechanism candidates (higher-Casimir + RS-coding + mixed); mass-mechanism heterogeneity per generation confirmed (gen-1 chirality + gen-2 Lorentz-integration + gen-3 higher-Casimir-or-Mersenne); multi-week explicit derivation distinguishes substrate-mechanism FORCING form.

v0.2 Substrate-Symplectic Cat 6 Cross-Verification for Gen-3 T2003 (~12:20 EDT)

Substrate-Symplectic Substrate-Pfaffian at V_(5/2, 1/2) Gen-3

Per Lyra Substrate-Symplectic Integer K-Types v0.1 (Thursday ~12:14 EDT) + Gate 4 v0.2 substrate-Pfaffian candidate: substrate-symplectic Cat 6 substrate-mechanism UNIVERSAL across substrate K-type spectrum.

Substantive cross-K-type verification target: does substrate-symplectic substrate-Pfaffian 1/g substrate-natural factor extend to V_(5/2, 1/2) gen-3 tau K-type? If so, gen-3 T2003 mass mechanism may also involve Cat 6 substrate-symplectic contribution.

Substrate-Mechanism Reading Update v0.2

Per Substrate-Symplectic Integer K-Types v0.1: V_(5/2, 1/2) carries substrate-symplectic content via Sp(2) $\cong$ Spin(5) accidental isomorphism. Substrate-Pfaffian Pf(ω) at V_(5/2, 1/2) is substrate-natural per Sp(2) substrate-symplectic structure.

Possible substrate-mechanism reading for T2003 = $g^2 \cdot$ (2^{C_2} + g):

If gen-3 mass-mechanism composition (analogous to Gate 4 v0.2 for gen-2):

$$\frac{m_\tau}{m_e} = \frac{\text{Pochhammer ratio gen-3/gen-1} \cdot \sqrt{\text{Casimir gen-3/gen-1}} \cdot W_\lambda^{\text{Lorentz}}}{Pf(\omega)_{gen-3}}$$

With: - Pochhammer ratio gen-3/gen-1 = n_C · 2^rank = 20 (Wednesday Elie 3742) - Casimir ratio C_(5/2, 1/2) / C_(1/2, 1/2) = 29/5 ≈ 5.8 → $\sqrt{}$(C ratio) ≈ 2.408 - W_λ^Lorentz: substrate-mechanism candidate for gen-3 substrate-Lorentz-integration weight

Without explicit substrate-symplectic substrate-Pfaffian at V_(5/2, 1/2), the composition is multi-week pending.

Substrate-Symplectic Cat 6 + Gen-3 Heterogeneity Cross-Link

Per Gate 10 v0.1: gen-3 T2003 substrate-mechanism is HETEROGENEOUS from gen-2 T190 — different operator structure ($g^2 \cdot$ (2^{C_2} + g) ≠ (24/π^2)^{C_2} raised to different power).

Substrate-mechanism candidate v0.2 (substrate-symplectic + heterogeneity): gen-3 substrate-mechanism may involve substrate-symplectic substrate-Pfaffian contribution AT DIFFERENT WEIGHT than gen-2: - Gen-2 V_(3/2, 1/2): Cat 6 substrate-Pfaffian factor 1/g (per Gate 4 v0.2) - Gen-3 V_(5/2, 1/2): Cat 6 substrate-Pfaffian factor (potentially different per K-type)

Substrate-mechanism for substrate-Pfaffian at V_(5/2, 1/2): multi-week explicit derivation per Sp(2) substrate-symplectic Sym^k fundamental representation theory.

Refined Three-Candidate Reading for Gen-3 (Gate 10 v0.2)

Per Thursday afternoon substrate-symplectic Cat 6 substantive observation:

Candidate 1 (substrate-higher-Casimir): T2003 via higher Casimir C_3, C_4 on V_(5/2, 1/2) per SSG-16 substrate-higher-Casimir Cat A

Candidate 2 (substrate-Mersenne RS coding): T2003 via SSG-8 + SSG-19 substrate-Mersenne / substrate-Affine RS coding on GF(2^g)

Candidate 3 (substrate-symplectic + higher-Casimir composite): T2003 via Cat 6 substrate-symplectic + Cat 2 higher-Casimir composite at V_(5/2, 1/2) per Sp(2) substrate-symplectic + higher Casimir substrate-Lie-algebra structure ★ NEW per Thursday afternoon

Candidate 4 (substrate-symplectic + Mersenne composite): T2003 via Cat 6 substrate-symplectic + Cat 4 substrate-Mersenne via Sp(2) ≅ Spin(5) ↔ Mersenne SSG-8 cross-link ★ NEW per Thursday afternoon

Substantive expansion from 3 candidates (Gate 10 v0.1) to 4 candidates per Thursday substrate-symplectic Cat 6 substrate-mechanism.

Multi-Week Explicit Substrate-Symplectic Pf(ω) at V_(5/2, 1/2)

Per Cal #189 question-shape: multi-week explicit substrate-symplectic Pf(ω) at substrate K-type V_(5/2, 1/2) (analogous to Step Gate-4-S1 for V_(3/2, 1/2)):

- Substrate Sp(2) higher representation theory: V_(5/2, 1/2) ↔ higher Sp(2) representation
- Substrate-Pfaffian Pf(ω) at V_(5/2, 1/2) per Sp(2) substrate-symplectic structure
- Substrate-mechanism contribution to T2003 composition

Closure v0.2

Gate 10 gen-3 T2003 substrate-mechanism v0.2 extends v0.1 candidate list with substrate-symplectic Cat 6 substrate-mechanism contributions:

- 4 substrate-mechanism candidates for T2003 substrate-mechanism FORCING form (1 substrate-higher-Casimir + 1 substrate-Mersenne + 2 NEW substrate-symplectic composite candidates per Thursday afternoon)
- Multi-week explicit substrate-symplectic Pf(ω) at V_(5/2, 1/2) gates substrate-mechanism FORCING form per Cal #189.

Tier: K3 v0.16 Gate #10 at FRAMEWORK candidate (expanded from 3 candidates to 4 with substrate-symplectic Cat 6 contributions); multi-week explicit derivation distinguishes substrate-mechanism FORCING form.

— Lyra, Thu 2026-06-04 ~12:20 EDT. Gate 10 v0.2: substrate-symplectic Cat 6 cross-K-type verification for V_(5/2, 1/2) gen-3 + 2 NEW substrate-symplectic composite candidates (Cat 6 + Cat 2 higher-Casimir; Cat 6 + Cat 4 substrate-Mersenne); 4 substrate-mechanism candidates for T2003 FORCING form; multi-week explicit substrate-Pfaffian at V_(5/2, 1/2).